PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONGIDO DISTRICT EXAMINATION CYNDICATE.

FORM THREE MID-TERM EXAMINATION MARCH-2021

CHEMISTRY MARKING SCHEME.

Qn1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	C	A	D	D	C	A	A	A	E

1 @ 10 = 10 marks.

Qn2.

LIST A	i	ii	iii	iv	v
LIST B	G	C	E	B	I

1 @ 05 = 05 marks.

Qn3 (a) Strength of an acid reffered to the degree of dissociation of acid when dissolve in water. (01)

While.

Concentration of an acid reffered to the amount of acid dissolved in a given volume of water. (01)

(b) four uses of salts.

(i) used as Inorganic fertilizers eg. $(NH_4)_2SO_4$, KNO_3 etc.
(ii) used to relieve stomach acidity and heartburn eg. $MgSO_4$.
(iii) used to improve taste of food eg. NaCl.
(iv) used as baking powder eg. $NaHCO_3$. 1 @ 4 = 04 marks.
(v) use for food preservation.

Qn4. (a) Hardness of water is beneficial because;
(i) taste better due to the dissolved minerals.
(ii) Contains dissolves Ca^{2+} responsible for strong teeth and bones.
(iii) important for formation of strong shell of aquatic organisms.
1 @ 03 = 03 marks.
(iv) formation of ~~to~~ scum prevent water links in pipelines.

(b) (a) Volume use is the same because ion exchanger soften both temporary and permanent hardness of water (01½)

(b) Sample 'C' Contain both permanent and temporary hardness of water because after boiling volume of soap was reduced. (01½)

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Therefore some of temporary hard water was soften by boiling.

(c) Sample 'D' behave as distilled water since it consumed small amount of soap before boiling and after passing through the ions exchanger machine. (0½)

(d) Sample A hardness is due to $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$ (0½)

Sample B hardness is due to $CaSO_4$, $MgSO_4$ and Chloride of metals. (0½)

(e) sample A.

$$Ca(HCO_3)_2\ (aq) \xrightarrow{\Delta} CaCO_3\ (s) + H_2O\ (l) + CO_2\ (g).$$ (0½)

Sample B

$$CaSO_4\ (aq) + Na_2CO_3\ (aq) \longrightarrow CaCO_3\ (s) + Na_2SO_4\ (aq).$$ (0½)

Qn5 (a)(i) Empirical formula is the simplest formula which show the percentage Composition of a compound by mass. (0½)

(ii) Molecular formula is the chemical formula which show the actual number and type of atoms in a compound. (0½)

(b)(i) % water of crystallization = $100 - (22.7\% + 11\% + 22.3\%)$.

$= 44\%$. (0½)

Element	Zn	S	O	H_2O
% Composition	22.7	11	22.3	44
Divide % Composition by R.A.M.	$\frac{22.7}{65}$	$\frac{11}{32}$	$\frac{22.3}{16}$	$\frac{44}{18}$ (0½)
	0.35	0.34	1.39	2.44
Divide by the smallest number ratio	$\frac{0.35}{0.34}$	$\frac{0.34}{0.34}$	$\frac{1.39}{0.34}$	$\frac{2.44}{0.34}$ (0½)
	= 1	= 1	= 4	= 7

Empirical formula = $ZnSO_4 \cdot 7H_2O$ (0½)

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(ii) Molar mass of a Compound = 2 × Vapour density. (01/2)

= 2 × 143·5

= 287.

Molecula formula = n (Empirical formula) (01/2)

287 = n($ZnSO_4 \cdot 7H_2O$)

287 = 287n

n = $\frac{287}{287}$ (01/2)

= 1.

Molecular formula = 1 ($ZnSO_4 \cdot 7H_2O$)

= $ZnSO_4 \cdot 7H_2O$ (01/2)

(c) (i) PCl_5 – Phosphorus pentachloride. (01/2)

(ii) $CaCl_2$ – Calcium Chloride. (01/2)

(iii) $NaHCO_3$ – Sodium hydrogencarbonate / Sodium bicarbonate (01/2)

(iv) N_2O_4 – Dinitrogen tetraoxide. (01/2)

Qn6. (a) Molecular equation is a chemical equation in which formulas of the compounds are written as though all substances exist as a molecules and not as dissociated ions (01/2)

While.

Ionic equation is a chemical equation in which ionic Compounds in aqueous or solution are written as dissociated ions. (01/2)

(b) (i) $CuSO_{4(aq)} + 2NaOH_{(aq)} \longrightarrow Cu(OH)_{2(s)} + Na_2SO_{4(aq)}$. (01/2)

(ii) $CaCO_{3(s)} + 2HCl_{(aq)} \longrightarrow CaCl_{2(aq)} + H_2O_{(l)} + CO_{2(g)}$ (01/2)

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(c) (i) $Cu^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} + 2Na^+_{(aq)} + 2OH^-_{(aq)} \longrightarrow Cu(OH)_{2(s)} + 2Na^+_{(aq)} + SO_4^{2-}{}_{(aq)}$ (01/2)

$Cu^{2+}_{(aq)} + 2OH^-_{(aq)} \longrightarrow Cu(OH)_{2(s)}$ (01/2)

(ii) $CaCO_{3(s)} + 2H^+_{(aq)} + 2Cl^-_{(aq)} \longrightarrow Ca^{2+} + 2Cl^-_{(aq)} + H_2O_{(l)} + CO_{2(g)}$ (01/2)

$CaCO_{3(s)} + 2H^+_{(aq)} \longrightarrow Ca^{2+}_{(aq)} + H_2O_{(l)} + CO_{2(g)}$ (01/2)

Qn7. (a)(i) An atom is the smallest particle of an element which has all the chemical properties of that element. (01/2)

or.

An atom is the smallest particle of an element which take part in the chemical reaction.

(ii) Atomic number is the number of proton in an atom (01/2)

(iii) Mass number is the total number of proton and neutron in an atom. (01/2)

(b) (i) Atom is no longer indivisible particle of an element due to the presence of sub-atomic particles.

(ii) Atom can be created or destroyed.

(iii) Atom of the same element are not identical due to the presence of isotopes.

1/2@3 = 01 1/2.

(iv) Atoms can combined in any ratio to form compounds or molecular compound.

(c) Electronic configuration of element J = 2:8:2.

(i) valency = 2. (01/2)

group = II (01/2)

period = 3 (01/2)

(ii) Element J is metal. (01/2)

(iii) When J react with oxygen it form ionic compound

- Soluble in water.
- Conduct electricity when in molten or solution form

1/2@2 = 01

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- Have high melting and boiling point
- They are crystalline solid at room temperature.

8. (a) Iron – Fe
Fluorine – F
Argon – Ar
Silver – Ag
Nitrogen – N

1/2 @ 5 = 02½.

(b) To avoid confusion for one letter to represent two or more element. 02½.

Qn9. (a) First Aid Kit is a small box containing items used in provision of first Aid. 01

(b) - Promote healing and Recovering.
- Give hope.
- Prevent pain and Suffering.

1 @ 02 = 02 mark.

(c) - Razorblade – for cutting skin and dressing materials.
- Safety pin – for cripping bandage.
- Thermometer – for taking body temperature
- painkiller (paracetamol) – Relieving pain.
- Soap – for washing hand.

1 @ 2 = 02 marks

Qn10. (a) Flame is a region of burning gas that produce heat and light. 01

(b) - Because Non-Luminous flame is the hotest flame.
- Because Non-Luminous flame doesnot produce soot.

01 @ 2 = 02

(c)

Physical Change	Chemical change
- No new substance is formed.	- New substance is formed.
- process is reversible	- The process is Irreversible.
~~- Involve physical change of matter~~	1 @ 2 = 02

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Q.11. (a) (i) Fuel is any chemical combustible material when burn in air produce large amount of heat energy that can be used economically for domestic and Industrial purposes. ~~01~~ (02)

(ii) Destructive distillation is the process where organic fuel such as wood, coal and oil shale burn in absence of air (oxygen) to produce usefull product such as Charcoal, Coke, and petrol. ~~02~~ (03)

(b) (i) In order to get rid of volatile matter which Lower energy value of Coal. ~~01~~ (02)

(ii) Inorder to avoid formation of carbondioxide which Lower the energy value of water gas. ~~01~~ (02)

Q.n12. (a) (i) – air has no any chemical formula.
~~(ii)~~ – Air has no fixed Composition of its Constituents.
(iii) – Can be separated by physical means.
– Involve physical Combination of gases in the atmosphere. 1@2 = 02

(ii) – water has a specific chemical formular.
– water involve Chemical Combination of oxygen and hydrogen
– water has fixed Composition of its Costituents.
– Cannot be separated by physical means. 1@2=02

(b) (i) Chemical change.
(ii) Physical Change.
(iii) Chemical Change.
(iv) Physical Change.
(v) Chemical Change.

$\frac{1}{2}$@ 5 = (2.5)

(c) (i) Combustion is a chemical process which involve burning of material in presence of oxygen

(ii) Rusting is a chemical process occur in iron or Steel when come into Contact with water and air.

(iii) Fire fighting is process of set off the harmful fire using a suitable fire extinguisher. 1@3 = 03

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(d)(i) Condition necessary for fire to start.

- Fuel
- Heat (Sufficient). 1@3 = 03
- Air (oxygen).

(ii)

Burning material	Class of fire	Recommended extinguisher
Combustible metals.	Class D	- Dry powder (01) - Foam extinguisher.
Liquefied gas	Class C	- Dry powder. (01) - carbondioxide.
Solid materials eg. papers.	Class A	- use water (01)

Qn13. (a)(i) Mole is amount of substance that Contain Avogadro's number (6.022×10^{23}). (01)

OR.

- Is amount of substance which contain as many elementary particles as there atoms in 12-gram of Carbon-12 Isotope.

(ii) Avogadro's Law state that "~~each~~ One mole of any substance Contain the same number of particles (atoms, molecules, ions). (01)

ie. 1 mole = 6.022×10^{23} particles.

(b) Data.

$n = 0.23$ moles.

$L_A = 6.022 \times 10^{23}$ particles/moles

$N = ?$ (01)

Recall.

$N = nL_A$. (01)

$= 0.23$ mol $\times 6.022 \times 10^{23}$ particles/mol

$= 1.385 \times 10^{23}$ particles. (01)

∴ There is 1.385×10^{23} particles in 0.23 mol of Calcium nitrate. (01)

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(ii) Data.

$N = 214$ molecules.

$L_A = 6.022 \times 10^{23}$ molecule/mol.

$n = ?$ (01)

Recall.

$n = N / L_A$.

$= \frac{214 \text{ molecules}}{6.022 \times 10^{23} \text{ molecule/mol}}$ (01)

$= 3.55 \times 10^{-23}$ mols

∴ There is 3.55×10^{23} mol in 214 molecules of hydrogen. (01)

(iii) Data.

given volume. $= 5.7 dm^3$

GMV $= 22.4 dm^3$

$n = ?$

$N = ?$ (01)

$L_A = 6.022 \times 10^{23}$

Recall.

$n = \frac{\text{given volume.}}{\text{GMV.}}$

$= \frac{5.7 dm^3}{22.4 dm^3/mol}$ (01)

$= 0.25$ mol.

$N = n \times L_A$.

$= 0.25 mol \times 6.022 \times 10^{23}$ molecules/mol

$= 1.53 \times 10^{23}$ molecules. (01)

∴ There is 1.53×10^{23} molecules of nitrogen gas in $5.7 dm^3$ of nitrogen gas. (01)

(c)(i) Molar mass is a mass of one mole of a compound. (01)

(ii) $RMM(Ca(NO_3)_2) = (40 \times 1) + (14 \times 2) + (16 \times 6)$.

$= 40 g/mol + 28 g/mol + 96 g/mol$.

$= 164 g/mol$. (01)

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4. (a)(i) Atomic radius is the distance from the centre of an atom to the outermost stable shell. (03)

(ii) Ionization energy is the energy required to remove electron completely from an atom or ion (03)

(iii) Electronegativity is the ability of an atom to attract electron toward itself. (03)

(b)(i) Atomic radius decrease from left to the right ~~of~~ across the periodic table. (02)

(ii) Ionization energy increase from left to right across the periodic table. (02)

(iii) Electronegativity increase from left to right across the periodic table. (02)

Question.	Total marks.	
1	~~10~~ 10	
2	~~05~~ 05	
3	~~06~~ 06	
4	~~10~~ 06½	
5	~~14~~ 07	
6	~~08~~ 04	
7	~~12~~ 06	
8	~~08~~ 05	
9	~~09~~ 05	
10	~~13~~ 05	
11	~~014~~ 10	
12	~~18~~ 15·5	
*13	~~15~~ 15	} option.
*14	~~15~~ 15	

100%

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